

Day 4 : GPUs and Colling Systems Theory

Task 1 : Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

GPU Identification

My laptop has two graphics processors:

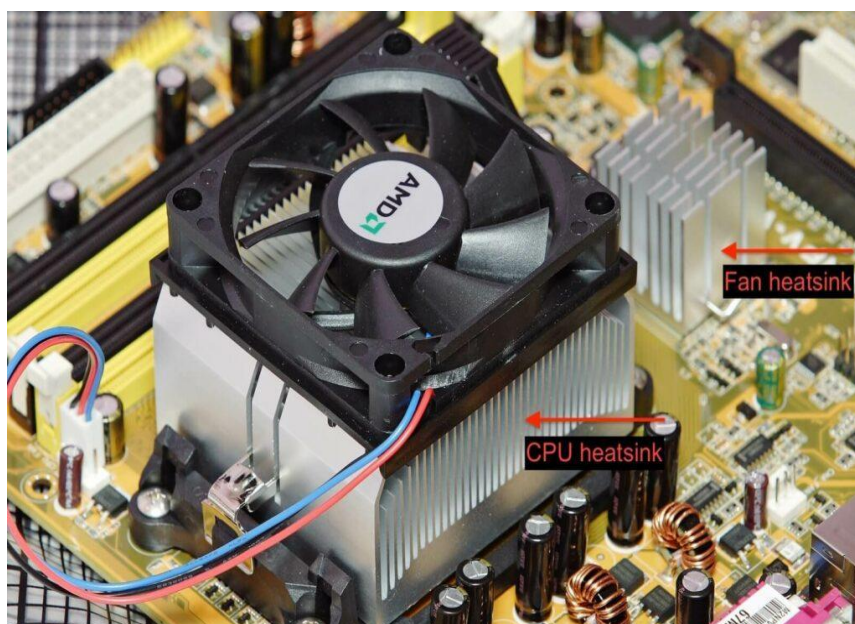
GPU	Brand	Model	Type
1	NVIDIA	GeForce RTX 3050 Laptop GPU (4 GB)	Dedicated GPU
2	Intel	Intel(R) UHD Graphics (128 MB)	Integrated GPU

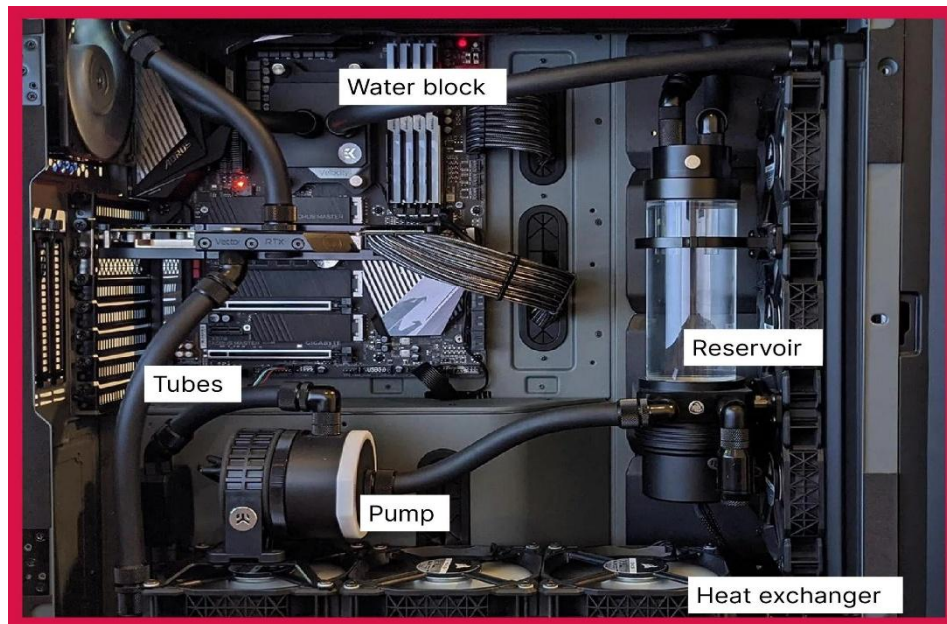
Explanation:

- The **NVIDIA GeForce RTX 3050 Laptop GPU** is a dedicated graphics card with **4 GB VRAM**, mainly used for gaming, graphics, and other demanding applications.
- The **Intel UHD Graphics** is an integrated GPU built into the processor and is mainly used for everyday tasks such as web browsing, videos, and basic applications.

My laptop uses both **integrated and dedicated graphics**, switching between them depending on the workload.

Task 2 : Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.





Cooling System Components

1. **CPU Fan** – The CPU fan moves air over the heatsink to remove heat from the CPU and prevent overheating.
2. **Heatsink** – The heatsink is attached to the CPU and absorbs the heat produced by it. It spreads the heat over a larger surface area so that it can be cooled by the fan.
3. **Case Fan** – The case fan maintains airflow inside the computer cabinet. It brings cool air into the case and removes hot air from inside.
4. **Thermal Paste** – Thermal paste is applied between the CPU and heatsink. It improves heat transfer from the CPU to the heatsink by filling tiny gaps between their surfaces.

Task 3 : Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	Air Cooling	Liquid Cooling
Cost	Generally cheaper and more affordable.	Generally more expensive due to the water block, pump, radiator, and other components.
Cooling Efficiency	Provides good cooling for most GPUs and normal workloads.	Usually provides better heat dissipation, especially under heavy workloads.
Maintenance	Requires low maintenance; mainly cleaning dust from fans and heatsinks.	Requires more maintenance and checking the pump, tubing, and coolant system.
Noise	Can be noisier because fans may run at high speed under heavy load.	Can be quieter in some setups, although it still uses fans and a pump.
Installation	Simple and easy to install.	More complex to install and requires additional space for the radiator and tubing.

Task 4 : Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

- I would choose **liquid cooling** for the GPU because I will use the PC for streaming IPL matches and playing graphics-heavy games.
- These activities can put a lot of load on the GPU and make it produce more heat.
- Liquid cooling can remove heat more effectively and keep the GPU temperature under control during long sessions.
- It also helps the GPU maintain stable performance and reduces the risk of overheating.
- Although liquid cooling is more expensive and needs more maintenance than air cooling, I think it is a better choice for a high-performance gaming and streaming PC.